Math-M-Addicts
MINDS THAT COUNT

SP250 Homework 5

1. Find the smallest positive integer which has all 10 digits in its decimal representation and is divisible by 72.

Solution:

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2. Heracles needs to fight the Lernean Hydra for one of his labors. A Hydra has many heads, and to defeat it, he must remove them all.

Heracles has two swords, and he can only use one of them at a time. The smaller sword will sever 21 of the Hydra's heads with a single stroke. The larger one will sever 100 heads with each stroke, but then the Hydra (unless it has just been killed) will grow 2023 new heads! If the Hydra initially has 101 heads, is it possible for Heracles to defeat it with his present swords or does he need to procure another sword?

NOTE: *A sword can only remove a particular number of heads, no fewer and no more.* If, for example, the Hydra has just 5 heads, they cannot be cut off with either of the swords.

Solution:

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3. Which number is larger: 2^{700} or 5^{300}?

Solution:

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4. The number of students in a certain classroom is between 30 and 40. It is known that each girl shook hands with exactly 3 boys but each boy shook hands with exactly 4 girls. How many students might be present in the classroom? Prove the uniqueness of your answer!

Solution:

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5. Prove that the only integer solution to the equation $3 \cdot x^2 = y^2 + z^2$ is $x = y = z = 0$.

Solution:

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6. There are 7 rooks on a 7×7 board so that none attacks another. Show that any 4×4 square contains at least 1 rook.

Solution:
